FPB-RA2E3 Bit-Banged UART TX Timing / Jitter Verification

Test Objective

Hardware timing verification was performed on the Renesas FPB-RA2E3 implementation of the transmit-only bit-banged UART following the UART jitter improvements.

The purpose of the test was to verify that the software UART meets the defined transmit timing requirements at 115200 baud.

UART Timing Requirements

For 115200 baud:

- Nominal bit period, Tbit = 1 / 115200 = 8.6806 µs
- Average baud-rate error requirement: ≤ ±1%
- TX bit-edge jitter design target: ≤ ±1% of Tbit
- TX bit-edge jitter absolute maximum: ≤ ±2% of Tbit
- ±1% jitter limit at 115200 baud: ±86.8 ns
- ±2% jitter limit at 115200 baud: ±173.6 ns
- Stop-bit duration requirement: ≥ 0.99 × Tbit
- TX request-to-start-edge latency is not included in the UART bit-jitter measurement.

For jitter measurement, the falling edge initiating a UART character is used as the timing reference. Corresponding transmitted edges are compared across repeated characters to determine the repeat-to-repeat edge timing variation.

Hardware Test

Platform: Renesas FPB-RA2E3
UART implementation: Bare-metal, software/bit-banged TX only
UART format: 115200 baud, 8 data bits, no parity, 1 stop bit (8N1)
Capture file: FPB-RA2E3_BitBangedUART_JitterTest_v09.csv

The captured data contained 158 complete UART characters distributed across five complete transmit bursts.

The captured frames decoded correctly as:

- [0032] Heartbeat OK
- [0033] Heartbeat OK
- [0034] Heartbeat OK
- [0035] Status message
- [0036] Heartbeat OK

All analyzed characters contained a valid HIGH stop-bit state and decoded correctly at 115200 baud.

Measured Baud Rate

The mean measured start-to-start character period was:

86.6617 µs

For an 8N1 character containing 10 bit periods, this corresponds to an effective baud rate of approximately:

115,391 baud

Baud-rate error relative to 115200 baud:

+0.166%

Requirement:

$\leq \pm 1\%$

Result: **PASS**

The measured baud-rate error is substantially inside the allowed ±1% tolerance.

Measured TX Edge Jitter

Corresponding UART transitions were compared at the same character/bit positions across repeated transmissions.

Worst measured deviation from the corresponding mean edge position:

±56.4 ns maximum

This is equivalent to:

0.65% of one nominal UART bit period

Measured peak-to-peak variation:

80 ns peak-to-peak

This is equivalent to approximately:

0.92% of one nominal UART bit period peak-to-peak

Acceptance target:

±1% Tbit = ±86.8 ns

Absolute maximum:

±2% Tbit = ±173.6 ns

Result: **PASS**

The maximum observed edge jitter of approximately 0.65% Tbit is below the ±1% design target and provides approximately 35% margin to the ±1% jitter limit.

The capture timestamps have approximately 40 ns timing granularity. The observed 80 ns peak-to-peak variation is therefore only approximately two capture timing increments, indicating that the actual software-generated jitter is approaching the resolution of the measurement capture.

Stop-Bit Verification

The minimum measured HIGH interval between the transition into the stop state and the next character start was:

9.72 µs

Nominal Tbit:

8.6806 µs

Minimum required stop interval:

0.99 × Tbit = 8.5938 µs

Measured minimum:

9.72 µs = approximately 1.12 × Tbit

Result: **PASS**

The transmitter therefore provides more than one complete nominal bit period of HIGH state before the following start bit.

Start-Bit Treatment

The falling edge that initiates each character establishes the UART timing reference (t0). Variation in when transmission is initially requested by software is considered TX-start latency rather than UART bit jitter and is therefore excluded from the jitter measurement.

Timing variation of subsequent edges associated with the start bit, data bits, and transition into the stop state is included in the corresponding-edge jitter analysis.

The implementation exhibits small deterministic, repeatable bit-position timing offsets. These offsets are distinct from jitter: jitter is the variation of a given edge from transmission to transmission. The repeat-to-repeat variation remains within the specified ±1% Tbit jitter target.

Verification Summary

Parameter	Requirement	Measured	Result
Nominal baud	115200 baud	115200 configured	—
Average baud error	≤ ±1%	+0.166%	**PASS**
TX edge jitter target	≤ ±1% Tbit (±86.8 ns)	±56.4 ns / 0.65% Tbit	**PASS**
TX edge jitter maximum	≤ ±2% Tbit (±173.6 ns)	±56.4 ns	**PASS**
Stop-bit minimum	≥ 0.99 Tbit / 8.5938 µs	9.72 µs / 1.12 Tbit	**PASS**
UART frame decode	Valid 115200 8N1	158 complete characters decoded successfully	**PASS**

Conclusion

FPB-RA2E3 BIT-BANGED UART TX TIMING/JITTER: PASS

Hardware measurement confirms that the FPB-RA2E3 transmit-only bit-banged UART meets the established UART TX jitter specification.

The measured maximum corresponding-edge jitter is approximately ±56 ns, or 0.65% of one nominal bit period, which is better than the ±1% design target and substantially below the ±2% absolute limit.

Average baud-rate accuracy is approximately +0.17%, also comfortably within the ±1% requirement. The minimum measured stop interval is approximately 1.12 nominal bit periods, providing adequate stop-bit margin.

The captured UART data decoded correctly at 115200 baud with valid stop states across all 158 complete analyzed characters.

Verification Status: PASS